

Data 605 Week 1 assignment 1

Keith DeNivo

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.1
v ggplot2    3.5.2      v tibble     3.2.1
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.0.4
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(RCurl)
```

Attaching package: 'RCurl'

The following object is masked from 'package:tidyr':

complete

```
library(MASS)
```

Warning: package 'MASS' was built under R version 4.5.2

Attaching package: 'MASS'

The following object is masked from 'package:dplyr':

select

```
library(lpSolve)
```

Warning: package 'lpSolve' was built under R version 4.5.2

```
#downloading Data
demandurl <- "https://raw.githubusercontent.com/division-zero/Data-605-Homework-KDD/refs/heads/main/demand.csv"
demandlink <- getURL(demandurl)
#read in the raw file
demand_df <- data.frame(read.csv(text = demandlink ))
#put the csv into a dataframe

head(demand_df)
```

	Region	Demand
1	Region1	2800
2	Region2	2400
3	Region3	2500

```
inventorylink <- getURL("https://raw.githubusercontent.com/division-zero/Data-605-Homework-KDD/refs/heads/main/inventory.csv")
#read in the raw file
inventory_df <- data.frame(read.csv(text = inventorylink ))
#put the csv into a dataframe

head(inventory_df)
```

	Warehouse	Region1	Region2	Region3
1	A	1200	900	800
2	B	1000	1100	700
3	C	900	700	1000

```
shipmentslink <- getURL("https://raw.githubusercontent.com/division-zero/Data-605-Homework-KDD/refs/heads/main/shipments.csv")
#read in the raw file
shipments_df <- data.frame(read.csv(text = shipmentslink ))
#put the csv into a dataframe

head(shipments_df)
```

	Warehouse	Region1	Region2	Region3
1	A	300	400	500
2	B	400	350	250
3	C	500	300	200

```
#head to view each dataframe
```

Scenario

A retail company operates three warehouses that distribute products to three sales regions. Management wants to evaluate current inventory levels, shipping patterns, and determine whether existing inventory can satisfy customer demand.

You have been provided the following datasets (files included as attachments; you need to access the assignment to download them):

inventory.csv

Warehouse,Region1,Region2,Region3

A,1200,900,800

B,1000,1100,700

C,900,700,1000

demand.csv

Region,Demand

Region1,2800

Region2,2400

Region3,2500

shipments.csv

Warehouse,Region1,Region2,Region3

A,300,400,500

B,400,350,250

C,500,300,200

Tasks

Part A: Inventory Analysis

Using vector operations:

1. Calculate total inventory available in each region.

```
#sum the values in the columns in inventory data frame and display.  
region1inventory <- c(inventory_df[, 2])  
#display region 1's inventory values  
print.listof(region1inventory)
```

```
Component 1 :  
[1] 1200
```

```
Component 2 :  
[1] 1000
```

```
Component 3 :  
[1] 900
```

```
#summing region 1 inventory  
cat("region1 sum: ", sum(region1inventory),'\n')
```

```
region1 sum: 3100
```

```
#repeat for region 2 and 3.  
region2inventory <- c(inventory_df[, 3])  
print.listof(region2inventory)
```

```
Component 1 :  
[1] 900
```

```
Component 2 :  
[1] 1100
```

```
Component 3 :  
[1] 700
```

```
cat("region2 sum: ", sum(region2inventory),'\n')
```

```
region2 sum: 2700
```

```
region3inventory <- c(inventory_df[, 4])
print.listof(region3inventory)
```

```
Component 1 :
[1] 800
```

```
Component 2 :
[1] 700
```

```
Component 3 :
[1] 1000
```

```
cat("region3 sum: ", sum(region3inventory),'\n')
```

```
region3 sum: 2500
```

1. Calculate inventory surplus or shortage relative to demand.

```
#extract the values of the demands for the regions from demand data frame
regionsdemand <- c(demand_df[,2])
print('each regions demand')
```

```
[1] "each regions demand"
```

```
print(regionsdemand)
```

```
[1] 2800 2400 2500
```

```
#put the sum of the inventory of the regions into a vector
inventorysum <- c(sum(region1inventory), sum(region2inventory), sum(region3inventory))
print(inventorysum)
```

```
[1] 3100 2700 2500
```

```
#calculate and display the difference between the inventory and the demand
inventoryminusdemand <- inventorysum - regionsdemand
cat("Surplus for each region: \n", "region 1: ", inventoryminusdemand[1],"\n", "region 2
```

```
Surplus for each region:
region 1: 300
region 2: 300
region 3: 0
```

Identify which region is most at risk of stock shortages.

Region 3 due to demand equaling inventory. region 1 and region 2 have 300 surplus each.

Interpret the business implications.

If demand fluctuates at all then region 3 will not be able to fulfill demand that is higher than the current demand. There will be missed sales in the region and possible customer dissatisfaction. It should be worth noting that having exactly what is needed when it is needed is the most efficient, however it is too risky.

Part B: Shipment Matrix Analysis

Using matrix operations:

1. Create a shipment matrix from the data.

```
head(shipments_df)
```

	Warehouse	Region1	Region2	Region3
1	A	300	400	500
2	B	400	350	250
3	C	500	300	200

```
#convert data frame to matrix  
shipments_matrix <- data.matrix(shipments_df)  
print(shipments_matrix)
```

	Warehouse	Region1	Region2	Region3
[1,]	1	300	400	500
[2,]	2	400	350	250
[3,]	3	500	300	200

1. Compute row totals and column totals.

```
shipments_df
```

	Warehouse	Region1	Region2	Region3
1	A	300	400	500
2	B	400	350	250
3	C	500	300	200

```
#calculating row total excluding first column because it is the warehouse label
rowtotal <- rowSums(shipments_matrix[,-1])

#displaying the row totals with the warehouse name
cat(colnames(shipments_df)[1], '\n',shipments_df[1,1], ' ',rowtotal[1],'\n',shipments_df[2,1], ' ',rowtotal[2],'\n',shipments_df[3,1], ' ',rowtotal[3],'\n')
```

```
Warehouse
A    1200
B    1000
C    1000
```

```
#the column totals excluding the warehouse label from total
columntotal <- colSums(shipments_matrix[,-1])
cat('column totals: ', '\n')
```

```
column totals:
```

```
print(columntotal)
```

```
Region1 Region2 Region3
1200    1050    950
```

Determine which warehouse ships the most products.

Warehouse A ships the most products at 1200.

Determine which region receives the most products.

Assuming the shipment csv shows warehouses shipping to regions, then region 1 receives the most products at 1200 products.

Part C: Resource Allocation Model

After reviewing inventory levels and shipment activity, management wants to determine how many additional truckloads of inventory should be sent from each warehouse next week to meet projected demand.

Let:

x = number of truckloads shipped from Warehouse A
y = number of truckloads shipped from Warehouse B
z = number of truckloads shipped from Warehouse C

Based on transportation capacity and regional replenishment requirements, management has developed the following system of equations:

$$x+y+z=18$$

$$2x+y=26$$

$$y+2z=22$$

These equations represent:

Total truckloads available for shipment.

Capacity constraints at Warehouse A.

Combined delivery requirements for Warehouses B and C.

Tasks

1. Express the System in Matrix Form

Rewrite the system as:

$$Ax=b$$

Identify:

The coefficient matrix A

The variable vector x

The constants vector b

```
#construct the matrix  
truckloadmatrix <- matrix(c(1,1,1,18,  
    2,1,0,26,  
    0,1,2,22), nrow = 3, byrow = TRUE)  
print('augmented matrix')
```

```
[1] "augmented matrix"
```

```
print(truckloadmatrix)
```

	[,1]	[,2]	[,3]	[,4]
[1,]	1	1	1	18
[2,]	2	1	0	26
[3,]	0	1	2	22

```
#constraints vector  
print('constraints/values b')
```

```
[1] "constraints/values b"
```



```
b <- c(truckloadmatrix[,4])
print(b)
```

```
[1] 18 26 22
```

```
#coefficent matrix
print('coefficent matrix')
```

```
[1] "coefficent matrix"
```

```
A <- truckloadmatrix[,-4]
print(A)
```

```
      [,1] [,2] [,3]
[1,]     1     1     1
[2,]     2     1     0
[3,]     0     1     2
```

Coefficient matrix $A = A$

variable vector $x = x$

constrains vector $b = b$

$$Ax = b \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 0 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 18 \\ 26 \\ 22 \end{bmatrix}$$

2. Solve the System Computationally

Using Python or R:

Create the coefficient matrix.

Create the constants vector.

Solve for x, y, and z.

Report the values obtained.

```
#x <- solve(A,b)
#print(x)
#Error in solve.default(A, b) :
# Lapack routine dgesv: system is exactly singular: U[3,3] = 0

#x <- qr.solve(A,b)
#print(x)
#Error in qr.solve(A, b) : singular matrix 'a' in solve

det(A)
```

```
[1] 0
```

```
print(det(A))
```

```
[1] 0
```

Since the determinant is zero that means the matrix is singular and non-invertible.

For a system of linear equations that means the equations are not linearly independent, so an infinite number or no solutions could exist. “solve” in R can not find a solution.

```
#to take a guess at a solution
#From Mass Package
#Calculates a Moore-Penrose generalized inverse or psuedoinverse of a matrix to approximate a
#x=A^-1*b
print('A^-1')
```

```
[1] "A^-1"
```

```
print(ginv(A))
```

```
      [,1]      [,2]      [,3]
[1,] 0.1111111 0.3611111 -0.1388889
[2,] 0.1111111 0.1111111 0.1111111
[3,] 0.1111111 -0.1388889 0.3611111
```

```
print('b')
```

```
[1] "b"
```

```
print(b)
```

```
[1] 18 26 22
```

```
x <- ginv(A) %*% b  
print('solved x with moore penrose generalized inverse:')
```

```
[1] "solved x with moore penrose generalized inverse:"
```

```
print(x)
```

```
      [,1]  
[1,] 8.333333  
[2,] 7.333333  
[3,] 6.333333
```

3. Verify the Solution

Verify that the solution satisfies all three equations by:

Substituting the values into each equation, or

Demonstrating computationally that:

$Ax=b$

Include your output.

```
print("Ax=b")
```

```
[1] "Ax=b"
```

```
print("A")
```

```
[1] "A"
```

```
print(A)
```

```
      [,1] [,2] [,3]  
[1,]    1    1    1  
[2,]    2    1    0  
[3,]    0    1    2
```

```
cat('\n', 'b:', b, '\n')
```

```
b: 18 26 22
```

```
bsolved <- A %*% c(x)  
print("approximated b")
```

```
[1] "approximated b"
```

```
print(bsolved)
```

```
      [,1]  
[1,]    22  
[2,]    24  
[3,]    20
```

```
print("original b")
```

```
[1] "original b"
```

```
print(b)
```

```
[1] 18 26 22
```

```
print("residuals")
```

```
[1] "residuals"
```

```
print(b-bsolved)
```

```
      [,1]  
[1,]   -4  
[2,]    2  
[3,]    2
```

The values generated $x = 8.33$, $y = 7.33$, $z = 6.33$ do not satisfy the equations.

They generate a constants vector 22, 24, 20 which gives residuals of -4, 2, 2 for 18, 26, 22.

```
print(" %error")
```

```
[1] " %error"
```

```
print('equation 1:')
```

```
[1] "equation 1:"
```

```
print(abs(-4)/18*100)
```

```
[1] 22.22222
```

```
print('equation 2:')
```

```
[1] "equation 2:"
```

```
print(abs(2)/26*100)
```

```
[1] 7.692308
```

```
print('equation 3:')
```

```
[1] "equation 3:"
```

```
print(abs(2)/22*100)
```

```
[1] 9.090909
```

we can not use this as a possible solution since it does not satisfy the physical “hard” constraints

number truckloads available, which is impossible to go above

$$x+y+z \leq 18$$

warehouse A capacity which is impossible to go above

$$2x+y \leq 26$$

Delivery requirements:

$$y+2z = 22$$

now is there a solution that meets the above requirements?

```
#using lpSolve to solve the above proposed equations
```

```
# equation to minimize x + y + z total trucks
```

```
minimize_eq <- c(1, 1, 1)
```

```
#Matrix
```

```
coeff_mat <- A
```

```
# =, <=, etc.
```

```
signs <- c("<=", "<=", "=")
```

```
# constraints
```

```
print("b")
```

```
[1] "b"
```

```
print(b)
```

```
[1] 18 26 22
```

```
# solving for the equations
solution <- lp("min", minimize_eq, coeff_mat, signs, b, all.int = TRUE)

print("solution")
```

```
[1] "solution"
```

```
print(solution)
```

Success: the objective function is 11

```
cat("Shipped from warehouse A x:", solution$solution[1], '\n')
```

Shipped from warehouse A x: 0

```
cat("Shipped from warehouse B y:", solution$solution[2], '\n')
```

Shipped from warehouse B y: 0

```
cat("Shipped from warehouse C z:", solution$solution[3], '\n')
```

Shipped from warehouse C z: 11

```
new_b <- A%%c(solution$solution[1],solution$solution[2],solution$solution[3])
print('residuals')
```

```
[1] "residuals"
```

```
print(b-new_b)
```

```
      [,1]
[1,]    7
[2,]   26
[3,]    0
```

Satisfies deliveries equation 1 residual 0, Does not use any warehouse A capacity equation 2 residual 26. uses 7 less truckloads than total.

4. Business Interpretation

Support your explanation using results from Parts A and B whenever appropriate.

Write a brief discussion (150–250 words) addressing:

How many truckloads should each warehouse ship?

To satisfy the deliveries Warehouse C can ship 11 truckloads. Warehouses A and B do not have to ship any. 11 trucks the the minimum number of truckloads shipped.

Which warehouse contributes the most inventory?

Warehouse C contributes the most inventory since every truckload from warehouse C counts as 2 deliveries warehouse B truckload only counts as 1 delivery. so if you want to use the least amount of truckloads you should use shipments from warehouse C.

How does the mathematical solution help management allocate resources efficiently?

Using this solution you do not need to use any warehouse A capacity needlessly. It uses the fewest number of truckloads to satisfy the delivery.

Why are systems of linear equations useful in supply chain planning and logistics?

Systems of linear equations are useful in supply chain planning and logistics because they allow you to optimize the resources you have in order to meet the physical constraints given such as number of raw materials, number of transportation vehicles, storage space, delivery demand etc. In the real world there may be more than one solution, but it usually best to find the minimum to reduce the resource consumption. If you can set up the equations you can find unique solutions that may satisfy the problem.